

5(a)

any **one** from:

provides energy for the, production / secretion, of milk ;
a substrate to form, milk sugar / lactose (in milk) ;
AVP ;

1

Question	Answer	Marks
5(b)(i)	(pale) blue lilac / mauve / purple / AW ; green / yellow / orange / red green / yellow / orange / red ;	2
5(b)(ii)	<i>any two from:</i> reducing sugar / glucose, diffused, into tubing or reducing sugar / glucose, moved into tubing, down a concentration gradient / from high to low concentration ; reducing sugar / glucose, small enough to move across tubing (so positive result in tubing and in beaker) ; hPRL / protein, too large to move into tubing (so only positive result in beaker) ; only movement of glucose was tested so no explanation possible for reducing sugars in general ;	2
5(c)	<i>any two from:</i> facilitated diffusion ; uses (transport / carrier / integral membrane) proteins ; specific / specific binding site ; conformational change / AW ;	2

Question	Answer	Marks
5(d)	any four from: 1 hPRL is cell signalling molecule ; 2 acts on target, cells / tissues or acts on cells with PRLR ; 3 PRLR is (cell surface membrane) receptor ; 4 hPRL binds to PRLR ; 5 complementary binding / ref. to specificity ; 6 leads to / sets off, (specific) responses in mammary gland cell or detail ; e.g. secondary messenger triggered enzymes activated	4

Question	Answer	Marks
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